[ENTERPRISE AUTONOMOUS QA PLATFORM] ● STATUS: PRODUCTION DEPLOYED (V2.4.0)

SilentQA Engine

System Architecture, Operational Overview & Complete User Manual

Executive Overview: SilentQA is a next-generation software quality assurance flight-recorder and autonomous defect intelligence engine. It captures high-fidelity browser micro-telemetry (DOM mutations, network HAR streams, user interactions, console faults, and rendering signals) in real time with zero perceived overhead. Equipped with built-in PII redaction (Luhn algorithm validation, cryptographic token masking), synthetic signal detectors (rage clicks, dead clicks, duplicate requests), and Google Gemini AI orchestration, SilentQA synthesizes fully reproducible Playwright scripts, interactive root cause analysis (RCA), and automated Jira/GitHub tickets within seconds of user complaint submission.

Production Domain (Live):	**https://silentai.dpdns.org**
Vercel Edge Gateway:	https://hhhcct.vercel.app
DNS & Edge Security:	Cloudflare Edge Network (javon.ns & tiffany.ns)
Source Repository:	github.com/fazil-dev-tech/silent-qa-hackathon (branch: master)
Target Frameworks:	Next.js 15 (Turbopack), React 19, Supabase Postgres, Chrome MV3
Verification Status:	**17/17 Unit Tests Passing • 0 TypeScript Diagnostic Errors**
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Table of Contents:

1. Executive Vision & System Topology	Page 2
2. Telemetry Taxonomy & Synthetic Signals Engine	Page 3
3. Chrome Extension (Flight Recorder) User Manual	Page 4
4. Web Command Center & Fleet Dashboard Operations	Page 5
5. AI Defect Intelligence, RCA & Playwright Synthesis	Page 6
6. Admin Governance, Deployment & REST API Reference	Page 7

1. Executive Vision & System Topology

Software testing traditionally suffers from the **'reproduction gap'**: developers waste 30-50% of debugging time trying to reproduce cryptic bug reports submitted with incomplete steps, missing environment state, or unhelpful static screenshots. SilentQA solves this by recording continuous micro-telemetry on client applications and packaging it into deterministic, machine-executable reproduction suites.

End-to-End Architectural Flow

[1] CLIENT TIER	[2] INGESTION TIER	[3] INTELLIGENCE TIER
Chrome MV3 Extension • Content Script DOM Observer • Injected Network Hook • Offscreen Audio/Video Capture • Local Ring Buffer Storage	Next.js 15 Edge Routes • Batch Telemetry Streaming • Session Lock & De-dupe • Dynamic PII Sanitizer • Cloudflare Global CDN	AI Core & Synthesis • Google Gemini 1.5 Pro • Synthetic Signal Engine • Playwright Synthesizer • Supabase Vector/JSON DB

Monorepo Structure & Package Responsibilities

Directory / Package	Role & Technical Stack	Key Responsibilities
`apps/web`	Next.js 15, React 19, Tailwind CSS, Lucide, Framer Motion	Web command center, fleet sessions viewer, AI defect report viewer, admin console, edge API routes.
`apps/extension`	Chrome Manifest V3, Webpack/TS, Offscreen API, Content Scripts	Flight recording engine, DOM mutation interception, network HAR tracking, selector picker, bug complaint submitter.
`packages/shared`	Isomorphic TypeScript Library, Zod Validation Schemas	Canonical types (TelemetryEvent, DefectReport), Luhn PII scrubber, synthetic signal detectors, Playwright code generator.
`supabase`	PostgreSQL Database, Row-Level Security (RLS)	Persistent flight sessions, telemetry logs, defect reports, tester profiles, audit logs.

Session State Synchronization & Fleet Consistency

To ensure rock-solid stability during continuous live testing, the SilentQA Web Dashboard utilizes a **functional state lock** and **localStorage persistence** (`sqa_selected_session_id`). Background polling occurs silently every 10 seconds without resetting user selections, spinning buttons, or disrupting active investigations.

2. Telemetry Taxonomy & Synthetic Signals

SilentQA ingests micro-events into a standardized `TelemetryEvent` data structure. Events are tagged with millisecond epoch timestamps, severity levels, deterministic fingerprints, and selector paths.

Category	Event Types	Description & Extraction Logic
User Interactions	`click, input, form_submit`	Captures targeted DOM elements, unique CSS selectors, sanitized keystroke lengths, and submission triggers.
Network & HAR	`network, websocket_in, websocket_out`	Captures HTTP method, status codes (2xx, 4xx, 5xx), response durations (ms), payload shapes, and socket frame drops.
DOM & Rendering	`dom, route_change, navigation`	Tracks page navigation, SPAs client-side routing, layout shifts, element insertions, and viewport resizing.
Runtime Faults	`error, warning, performance`	Traps unhandled JavaScript exceptions, window error events, rejected promises, console.warn traces, and slow frames.

Synthetic Anomaly Detection Algorithms

SilentQA's signal engine analyzes temporal clusters of raw telemetry to synthesize high-confidence heuristic anomalies:

Signal Name	Trigger Heuristic & Window	QA Impact & RCA Implication
Dead Click	User clicks an element; 0 DOM mutations and 0 network requests occur within 1,000ms.	Indicates broken event listeners, missing click handlers, or unresponsive UI components.
Rage Click	≥3 consecutive clicks on the same DOM selector within a 500ms sliding window.	Highlights severe user frustration, frozen checkout buttons, or stalled asynchronous states.
Duplicate Request	Identical HTTP method + URL + payload fingerprint triggered within 1,000ms.	Detects missing button debouncing, double-charge billing hazards, and race conditions.
Slow Request	HTTP/API response duration exceeds 2,000ms (or customizable SLA threshold).	Flags backend microservice bottlenecks, unindexed DB queries, or edge timeouts.

Enterprise PII Redaction & Secret Masking

To satisfy GDPR, CCPA, and SOC 2 requirements, all telemetry passes through the client-side **SilentQA Scrubber** before persisting to disk or transmitting across network boundaries:

- **Luhn Algorithm Validation:** Mathematically validates and redacts Visa, Mastercard, AMEX, and Discover numbers (`[CARD_REDACTED]`).
- **Cryptographic Secret Masking:** Automatically identifies and masks Bearer Tokens, JWTs, API Keys, Passwords, and Session Nonces (`[SECRET_REDACTED]`).
- **National ID & Identity Redaction:** Sanitizes Indian PAN, Aadhaar, US SSN, Email Addresses, and Phone Numbers.

3. Chrome Extension (Flight Recorder) User Manual

The SilentQA Chrome Extension (`apps/extension`) operates silently in the tester's browser. It acts as an automated flight data recorder, continuously streaming events to the fleet telemetry buffer.

Feature / Control	Badge / State	Operational Function & Usage
Recording Toggle	**[RECORDING]** **[PAUSED]**	Starts or pauses telemetry recording. When active, a pulsing green radar dot displays. In paused state, an amber banner alerts the user that recording is suspended.
Station & Tester Tag	**[QA-STATION]**	Binds the flight session to a specific tester identity and hardware workstation (e.g. Chrome/Win11) for precise team telemetry attribution.
Target Host Indicator	**[ACTIVE HOST]**	Displays the active domain under inspection (e.g. `app.local` or `staging.ecommerce.com`) with auto-filtering.
Target Selector Picker	**[PICK ELEMENT]**	Activates an interactive crosshair overlay on the page. Clicking any element automatically computes its exact CSS selector and binds it to the complaint.
Complaint Window Slider	**15s / 30s / 60s**	Defines the retro-temporal capture window. When filing a bug, SilentQA packages all telemetry from the preceding N seconds leading up to the failure.
AI Defect Generator	**[GENERATE REPORT]**	Instantly bundles local ring-buffer events, scrubs sensitive tokens, sends the payload to the Next.js API, and opens the synthesized report.

Step-by-Step Testing Workflow

Step 1: Install & Pin the Extension: Open Chrome → `chrome://extensions` → Toggle 'Developer mode' → Click 'Load unpacked' → Select `apps/extension/dist`.

Step 2: Start Flight Recording: Click the SilentQA extension icon in the toolbar. Verify the green pulsing indicator displays '[RECORDING]'.

Step 3: Perform QA Testing: Browse the application under test naturally (fill forms, click buttons, navigate pages). SilentQA runs in the background with zero lag.

Step 4: Tag a Defect in Real Time: When a bug occurs (e.g. button unresponsive, unexpected error popup, broken checkout):

a) Click the extension popup or press hotkey `Alt+Shift+Q`.
b) Enter a quick note (e.g. *'Place Order button spins infinitely'*).
c) Optional: Click **[Pick Element]** to highlight the failing component.
d) Click **[Generate AI Report]**.

Step 5: View Synthesized Report: The extension automatically syncs with the web command center and opens the complete reproduction report.

4. Web Command Center & Dashboard Operations

The SilentQA Web Dashboard (`https://silentai.dpdns.org/app`) provides real-time fleet observability, session exploration, visual anomaly overlays, and report generation.

Dashboard Component	Control / Badge	Operational Function
Fleet Session Selector	**[DROPDOWN DRAWER]**	Lists all recorded flight sessions across the fleet. Shows session ID, host URL, tester tag, event count, and timestamp. Features persistent selection lock.
Fleet KPI Summary Strip	**[METRICS RIBBON]**	Displays 4 live KPIs: Total Fleet Sessions, Active Anomaly Rate (%), Critical Defect Count, and Mean Time to Resolution (MTTR).
Session Timeline Scrubber	**[TIMELINE SLIDER]**	Visual timeline mapping every micro-event chronologically. Click or drag to scrub session state backwards and forwards in time.
Live Telemetry Stream	**[EVENT LOG TABLE]**	Color-coded event log with badges for network requests, DOM mutations, console logs, and errors. Filterable by event type and severity.
Anomaly Heatmap Overlay	**[VIEWPORT OVERLAY]**	Projects bounding boxes over the application UI: Red (Critical Errors), Amber (Rage/Dead Clicks), Green (Successful User Actions).
Defect Reports Drawer	**[GENERATED REPORTS]**	Lists all AI-synthesized defect reports with severity pills, confidence scores, and one-click access to detailed inspection pages.

Navigating the Flight Session Viewer

1. **Selecting a Session:** Open the dropdown in the top-left corner. Pick any active or historical session. Notice that background fleet auto-syncing preserves your selection without resetting.
2. **Filtering Telemetry:** Use the top filter pills (`All`, `Errors Only`, `Network Only`, `Interactions`) to isolate relevant events.
3. **Inspecting Event Payloads:** Click any event row to expand its JSON metadata, request headers, payload shape, and stack trace.
4. **Exporting Fleet Telemetry:** Use the 'Export JSON' button to download complete sanitized session telemetry for offline analysis or bug archiving.

5. AI Defect Intelligence, RCA & Playwright Synthesis

When an issue is flagged, SilentQA executes a multi-stage AI reasoning pipeline powered by Google Gemini. The model analyzes the temporal sequence of user actions, DOM alterations, and network transactions to produce a comprehensive defect dossier.

Report Component	AI Generation Logic	Developer Value
Executive Title & Severity	Synthesizes concise, actionable title; assigns severity (Critical, High, Medium, Low).	Enables immediate triage in sprint planning and triage standups.
Reproducible Steps	Generates numbered, human-readable steps from user click and input telemetry.	Eliminates vague bug reports like 'checkout is broken'; gives exact sequence.
Root Cause Hypothesis (RCA)	Cross-correlates console errors, 4xx/5xx responses, and code stack traces.	Points developers directly to the offending file, function, or API endpoint.
Playwright Script	Synthesizes end-to-end executable TypeScript test reproducing the bug.	Instant regression test suite additions; runnable in CI/CD pipelines.
Suggested Fix & Checks	Recommends targeted code corrections, defensive checks, and unit tests.	Dramatically accelerates MTTR (Mean Time to Resolution).

Synthesized Playwright Test Reproduction Example

```
import { test, expect } from '@playwright/test';

test('reproduce: unresponsive checkout payment submission', async ({ page }) => {
  // 1. Navigate to target application
  await page.goto('https://app.local/checkout');

  // 2. Replay recorded user interactions
  await page.fill('#billing-zip', '94107');
  await page.click('button[type="submit"]#place-order');

  // 3. Automated Assertion (Verifying Anomaly)
  // Expected: HTTP 200 POST /api/checkout
  // Actual: Trapped 3 consecutive rage clicks & uncaught TypeError
  const errorBanner = page.locator('.error-notification');
  await expect(errorBanner).toBeVisible();
});
```

One-Click Enterprise Issue Export (Jira & GitHub)

SilentQA includes native integrations with Jira Cloud and GitHub Issues. From the report view, clicking **'Export to Jira'** or **'Create GitHub Issue'** automatically formats the title, severity, reproduction steps, environment metadata, and Playwright code block directly into the target tracker.

6. Admin Governance & Enterprise API Reference

The Administration Console (`/admin`) provides fleet control, security policies, and integration credential management.

Governance Domain	Settings & Capabilities	Enterprise Policy
Fleet Ingestion Rules	Rate limits per station, max event buffer size (default: 5,000 events/session), batch compression.	Prevents network congestion and database flooding during intensive exploratory testing.
PII Scrubbing Policy	Toggle rules: Credit Card (Luhn), Bearer Tokens, JWTs, Email, Phone, Custom Regex patterns.	Mandatory client-side enforcement guarantees zero sensitive customer PII persists to logs.
Data Retention & TTL	Configurable session retention (7 days, 30 days, 90 days), automated cold storage tiering.	Compliance with enterprise data governance and storage cost optimization.
Integrations Hub	Jira Cloud API tokens, GitHub Personal Access Tokens, Slack Webhook URLs, Gemini API Keys.	Centralized credential encryption with automated permission scoping.

Complete REST API Endpoint Reference

Method & Route	Payload / Parameters	Description & Return Type
`POST /api/telemetry`	`{ session_id, events: TelemetryEvent[] }`	Batch ingests telemetry events from Chrome extension into persistent storage.
`GET /api/fleet/sessions`	`?status=active&limit;=50`	Returns active and historical flight sessions with event counts and metadata.
`POST /api/report`	`{ complaint, session_id, window_sec }`	Executes Gemini AI prompt pipeline; returns synthesized DefectReport object.
`GET /api/sessions/:id/anomalies`	`URL param: session id`	Computes synthetic signals (rage clicks, dead clicks, duplicates) for session.

Cloudflare Edge Network & Custom Domain Architecture

SilentQA is deployed globally on Vercel Edge Infrastructure fronted by the Cloudflare Edge Network. Authoritative nameservers (`javon.ns.cloudflare.com` & `tiffany.ns.cloudflare.com`) provide CNAME flattening to `cname.vercel-dns.com` with automated HTTP-01 SSL verification, giving sub-50ms latency across global QA testing fleets.

Document Signoff: Verified against monorepo build, automated test suite (17/17 passing), and live production endpoint at **https://silentai.dpdns.org**. Approved for enterprise deployment.